ANTICYTHERE3D

Antikythera Mechanism simulator

User manual ▪ version 1.4.0



The earliest known astronomical calculator, in working order:
33 gears ▪ 15 arbors ▪ 2,231 teeth, a single crank.

Exact ratios after Freeth *et al.*, *Nature* 2006 & 2008, *Scientific Reports* 2021

Windows ▪ macOS ▪ Linux

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1 Installing the program

1.1 The ready-made package

Packages are on the repository's **Releases** page (`github.com/ARP273-ROSE/Anticythere3D/releases`). Each one **carries its own Python**: nothing else to install, and no administrator password is asked for.

System	File	Installation
Windows	`Anticythere3D-Setup.exe`	run the installer; it installs into your profile (`%LOCALAPPDATA%\Programs`), without administrator rights
macOS Apple Silicon / Intel	`Anticythere3D-macos-arm64.dmg` `Anticythere3D-macos-x86_64.dmg`	open the disk image, drag the application into **Applications**
Linux x86_64 / ARM64	`Anticythere3D-linux-x86_64.tar.gz` `Anticythere3D-linux-arm64.tar.gz`	unpack, run `./installer.sh`: installs into `~/.local/share`, adds a menu entry, **no sudo**

These names without a version number are permanent links to the latest release; the Releases page also lists the same files with their version number, and a `SHA256SUMS` file giving the fingerprint of each one.

Unsigned packages

Windows: "Windows protected your PC" — click **More info**, then **Run anyway**, once. **macOS**: the first time, **right-click** the application → **Open**, then confirm; if macOS still refuses: **System Settings** → **Privacy & Security** → **Open Anyway**.

1.2 Updates install themselves

On all three systems, a few seconds after opening, the installed program checks whether a newer release has been published. If so it offers it, with the release notes in the interface language; you can also ask with **Help** → **Check for updates…**. Only a **small code archive** is downloaded; its size and fingerprint are checked, it is unpacked in place of the old one, and the program restarts. When run from source, it replaces nothing: it offers the download page (or `git pull`).

If the update fails

The message "an application file is in use by another program" means another copy of the program is running. Close it (or restart the computer) and ask for the update again: the installed version is left untouched.

1.3 Incident reports

If it crashes, freezes or fails to start, the program can tell its maintainer by itself. **Nothing is sent without your consent**: the question is asked at first launch, and the **Help → Send crash reports automatically** option lets you change your mind. **Help → Report a problem...** sends a written report, in your own words. A report holds the version, the system and the error trace; no files, no names, no personal paths. Without a network, it waits for the next start.

1.4 Where your data lives

Settings, reports and log are **never** stored in the program folder, since an update replaces it entirely:

System	Data folder
Windows	`%LOCALAPPDATA%\Anticythere3D`
macOS	`~/Library/Application Support/Anticythere3D`
Linux	`~/.local/share/Anticythere3D` (or `$XDG_DATA_HOME`)

1.5 From source

All you need is **Python 3.10 or newer** (on Windows, double-clicking `launch.bat` does everything). Everything else — 3D included — is installed by the program itself on first start: `python run.py`.

What it installs, and where

Four packages: `PyQt6` (the interface), `numpy` (the geometry), `pyqtgraph` and `PyOpenGL` (the 3D view). Outside a virtual environment it installs with `--user`: **nothing is written into the system's packages**. On Linux, if Qt lacks system libraries, the program prints the exact command to run.

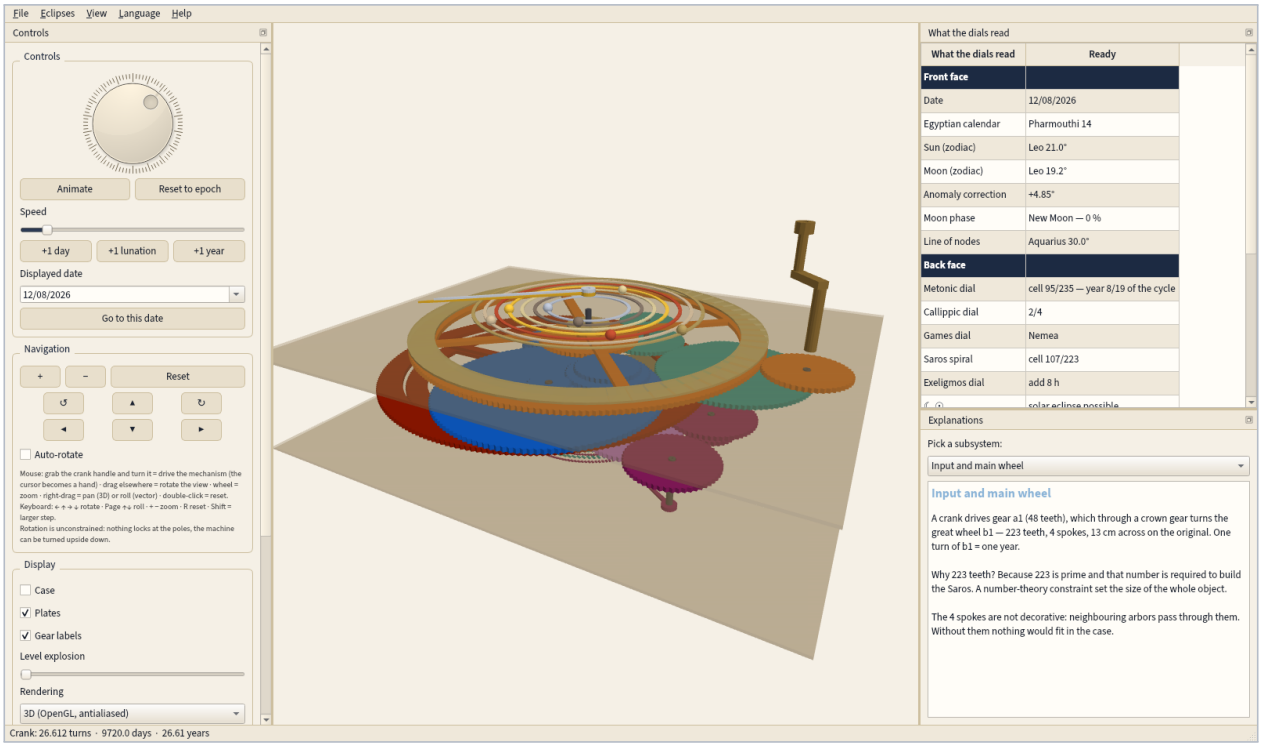
Command	Effect
`python run.py --check`	checks dependencies, style, logo and manuals, then exits
`python run.py --no-install`	starts without installing anything
`python run.py --lang fr`	French interface (default: the system's language)
`python run.py --vector`	starts in vector rendering
`python run.py --samples 16`	stronger antialiasing (default: 8)

If the 3D view does not start

On a machine without an OpenGL driver (server, virtual machine, remote desktop), **the program does not crash**: it switches by itself to *vector rendering*, which shows the same motions with no graphics dependency — and exports to SVG and PDF.

2 Getting started in two minutes

1. Open the program. The machine appears **closed**, as you would find it on a shelf: wooden case, engraved bronze front, hands, crank on the side. The program speaks the system's language; the **Language** menu lets you pick English or French, and the choice is remembered.
2. On first launch, one question: allow automatic incident reports or not (see above).
3. Drive the machine: **grab the crank handle** with the mouse, on the side, and turn it (the cursor becomes a hand) — or turn the **round knob** at the top left, one full turn of which is one year.
4. Untick **Case**: the case disappears and the 33 gears show up, turning.
5. Look at the right-hand panel: it permanently shows what the machine's owner read.



The case removed, on 12 August 2026: the 33 gears, the 15 arbors and the two large spoked wheels (b_1 at the front, e_3 at the back). On the right, the Moon is new and the panel announces a possible solar eclipse — the one of that very day.

A look that does not change from one computer to another

The program imposes its own style (parchment and night-blue colours, the Source Sans 3 font shipped with it): it looks the same on Windows, macOS and Linux, whether the system uses a light or a dark theme. Only the screen scaling (125%, 150%…) remains the system's.

3 The interface

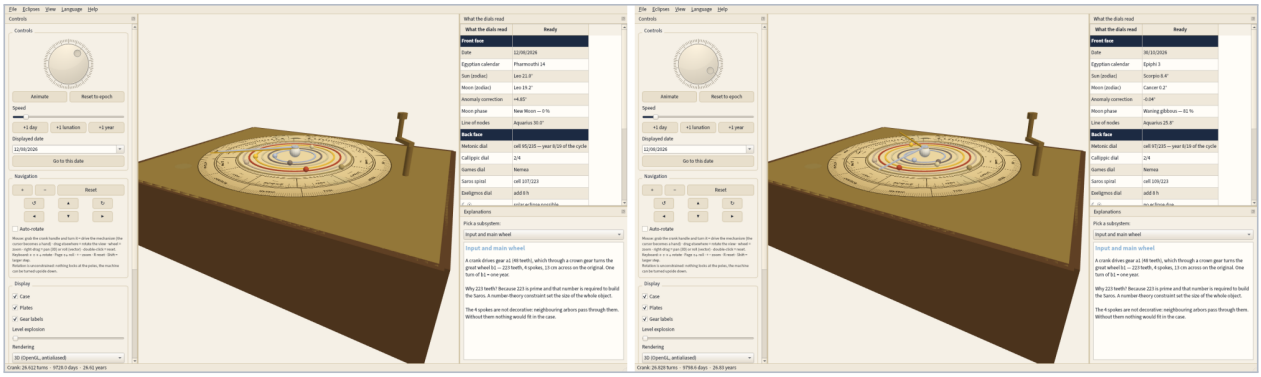
3.1 Controls

Item	Role
Crank handle	in the view: the machine's **only** input, grabbed with the mouse (see below)
Round knob	the same crank, geared down: one turn = one year
Animate / Pause	turns the crank continuously
Speed	simulated days per second
+1 day / +1 lunation / +1 year	advances by an exact step
Reset to epoch	brings the machine back to its calibration date
Displayed date + Go to this date	jumps straight to a date

Why a single input?

Because the real machine has only one. Everything else — Sun, Moon, phase, eclipses, calendars — is not "computed" separately: they are gear ratios, hence mechanical consequences of the crank's single motion. That is exactly what the program does: no display formula, only fractions of tooth counts.

3.2 Turning the crank with the mouse



Before and after one turn of the handle: the date moves from 12/08 to 30/10/2026 (78.6 days), the Moon from new to waning gibbous.

1. Move the mouse onto the crank's **handle** (or its arm), on the right-hand side of the case: the cursor becomes an **open hand**.
2. Press and hold the button: the hand closes, the handle is grabbed.
3. Move in a circle around the crank's axis: the machine moves forwards one way, backwards the other. Every dial follows, and so does the round knob.

How much is one turn of the handle?

The crank carries wheel a_1 (48 teeth), which meshes with the great wheel b_1 (223 teeth). One turn of the handle therefore turns b_1 by 48/223 of a turn — and b_1 turns once a year:

$$1 \text{ turn of the handle} = \frac{48}{223} \text{ year} \approx \frac{48}{223} \times 365.24 \approx 78.6 \text{ days},$$

in the opposite direction to b_1, as with any two meshing wheels. It takes $223/48 \approx 4.65$ turns of the handle for one year.

The gesture is the same in **vector rendering**, where the crank is drawn on wheel a_1. Away from the handle, dragging always rotates the view: you never move the machine by accident. The mouse wheel turned *over the round knob* also drives the machine.

3.3 Navigation: zoom, rotate, turn over

Gesture	Effect
Drag the handle	turn the crank (hand cursor)
Drag elsewhere	rotate the object
Mouse wheel	zoom in forwards, out backwards (bounded: you never go through the machine)
Right-drag	pan (3D) or roll (vector)
Double-click	reset the view
← ↑ → ↓	rotate with the keyboard
Page ↑ / Page ↓	roll (third axis)
+ / −	zoom in / out
R	reset the view
Shift held	larger rotation step
Space	animate / pause
F / B / I	front face / back face / three-quarter view
E	list of upcoming eclipses
F1	this manual

Truly free rotation

The camera is driven by a *quaternion*, not by Euler angles: **nothing locks at the poles**. You can go over the top, carry on, and turn the machine completely upside down. **Reset** (or R) always brings back exactly the same three-quarter view, whatever rolls came before.

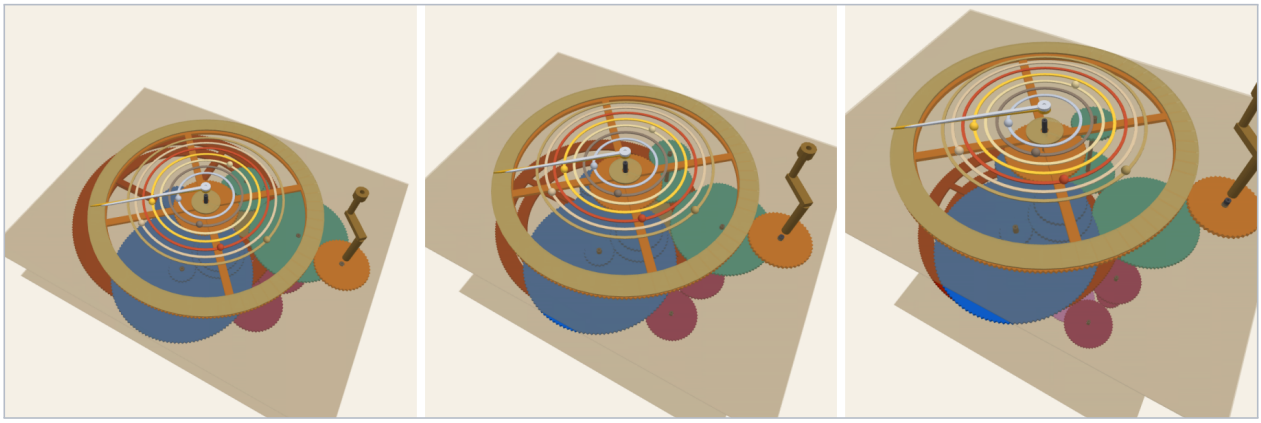
3.4 Display

Setting	Effect
Case	shows or removes the case and the front (the crank stays, so it can be grabbed)
Plates	the two inner plates that carry the arbors
Gear labels	labels each gear with its tooth count
Level explosion	spreads the 16 meshing planes apart along the arbors
Rendering	real-time 3D, or exportable vector
Tooth profile	triangular (ancient) or involute (modern)
Highlight	lights up one subsystem only, fades the others

The two tooth profiles

The teeth of the real machine are **nearly equilateral triangles**. The *mean* ratio is then strictly exact — which is why the instrument is right — but the *instantaneous* ratio wobbles as each tooth passes. The **involute of a circle**, unknown to the Greeks, is the only profile giving a constant ratio at every instant.

3.5 Level explosion



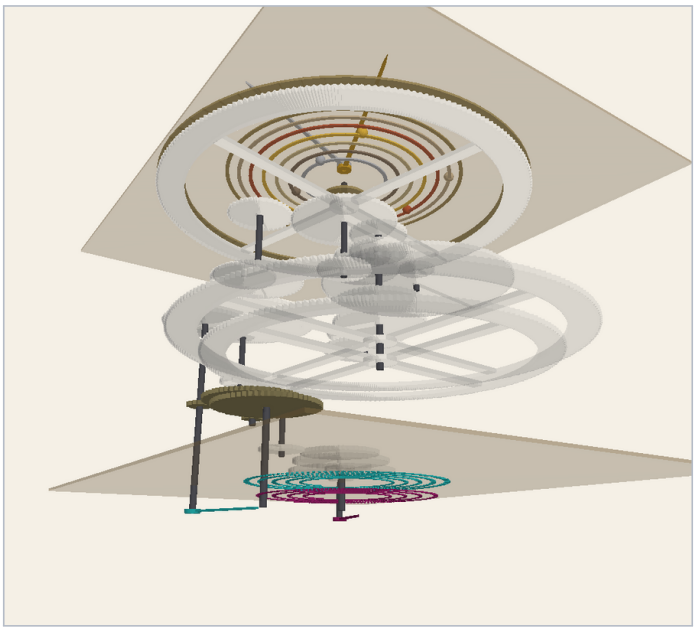
Explosion at 0, 50 and 100%: the stack stretches along the arbors, no gear leaves its plane.

The 33 gears are arranged on **16 levels**: two gears on the same level lie in the same plane and mesh. The **Level explosion** slider widens the gap between two levels from 3.4 mm (assembled machine) to 15.4 mm:

- the **back** stays in place and the stack stretches forwards; the front (dials, Cosmos, hands, crank) moves with the first level;
- **no gear moves within its plane**; meshing gears stay face to face, and the order of the levels never changes;
- the **arbors lengthen** with the stack: no gear floats;
- in vector rendering the slider is greyed out: the explosion would happen along the line of sight.

No part goes through another

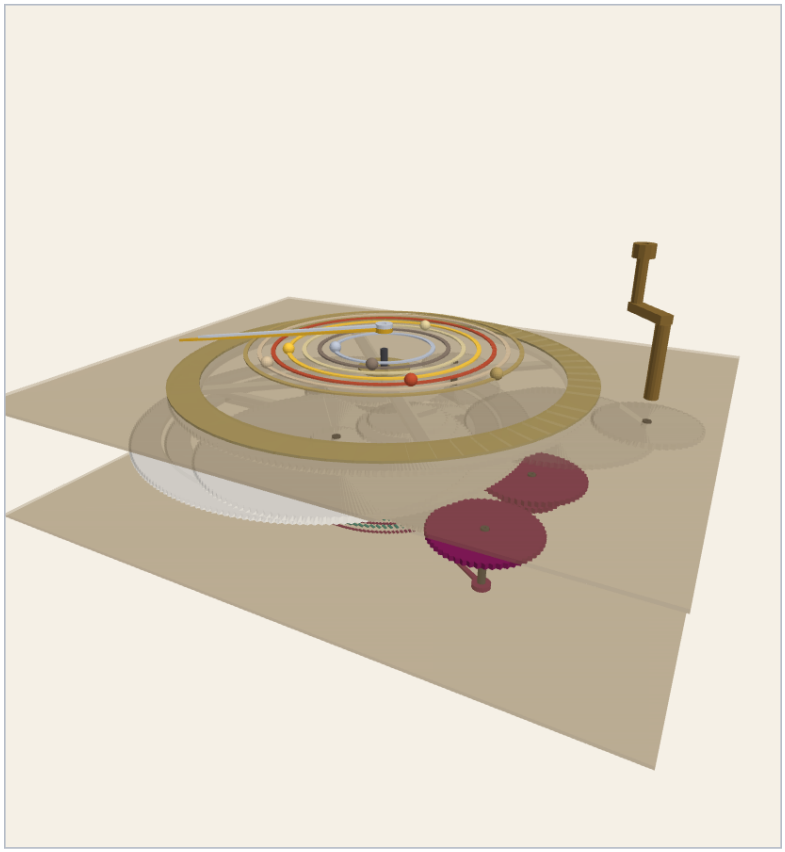
The test suite checks, at six slider positions and for 76 angles of the carrier e_3, that no gear enters another and that no arbor, **extended to its dial**, passes through a gear. This check led to two fixes in version 1.3.1: arbor o of the Callippic hand went through wheel i_1 of the Exeligmos (it was moved along its pitch circle around p, keeping the exact centre distance); wheels k_1, k_2 of the pin and slot, which turn with e_3, swept through arbors b and l (they now sit just in front of e_3, which carries them).



The Callippic train highlighted (exploded machine, three-quarter back view): arbor o runs down to its dial without going through any gear.

3.6 The colours of the seven trains

Each subsystem has its colour, in the 3D view, in the vector rendering and its exports, and in the legend. **Highlight** lights up just one:



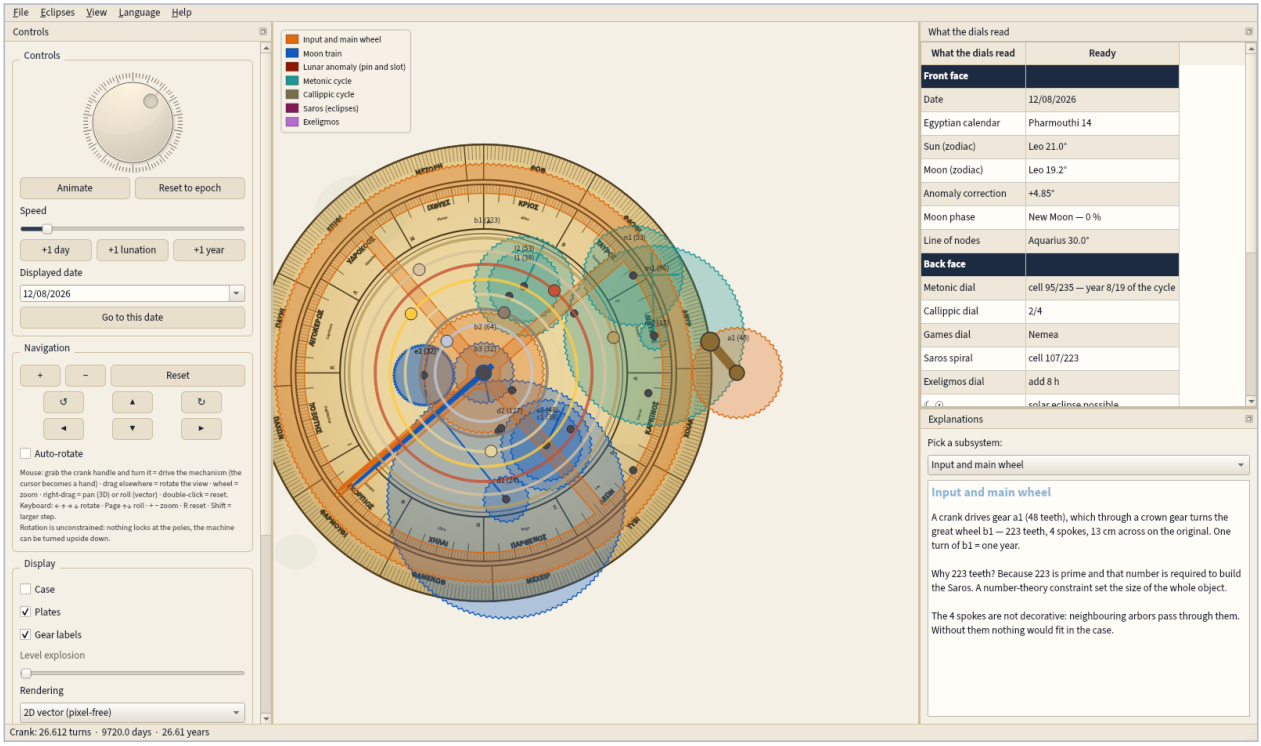
The Saros train highlighted, the others faded.

Colour	Code	Train	Gears
orange	`#DE6910`	Input and main wheel	a_1, b_1, b_2
blue	`#0D59C1`	Moon train	b_3, c_1, c_2, d_1, d_2, e_1, e_2, e_5, e_6
blood red	`#8E1700`	Lunar anomaly (pin and slot)	e_3, e_4, k_1, k_2
teal	`#1D9594`	Metonic cycle	l_1, l_2, m_1, m_2, n_1
khaki	`#786F4C`	Callippic cycle	n_2, p_1, p_2, o_1
raspberry	`#841959`	Saros (eclipses)	m_3, f_1, f_2, g_1
lilac	`#B66CCD`	Exeligmos	g_2, h_1, h_2, i_1

A computed palette, not picked by eye

The seven hues come out of an optimisation (`anticythere/palette.py`): a CIEDE2000 colour difference of at least 27.7 between all of them with normal vision; under simulated deuteranopia and protanopia, at least 19 between touching trains and 12 between all; a contrast of at least 3:1 on the parchment background. The tests check these thresholds at every release.

3.7 Vector rendering and exports

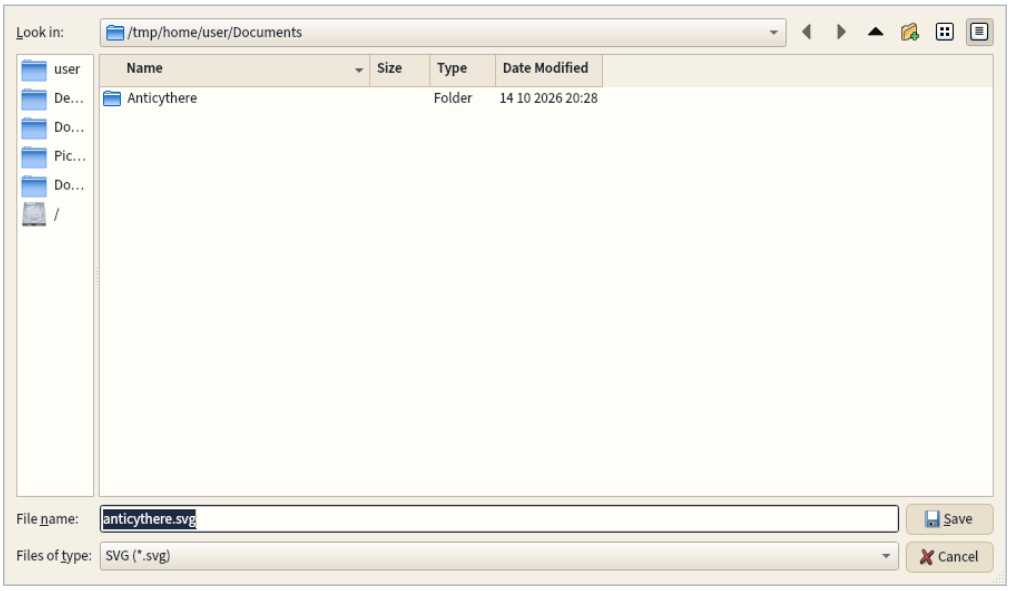


Vector rendering of the front face, with its legend: every tooth is a curve, not a pixel.

This mode draws the **real tooth outlines** as curves. It serves three purposes:

- looking at the mechanism flat, with nothing hidden;
- exporting to **SVG** or **A3 PDF** (**File**) — the files contain no bitmap image: print at A0 or zoom forever;
- working on a machine without OpenGL.

The **File** menu also offers a PNG screenshot of the window, a CSV export of the dial panel, and an **STL** export of the 33 gears, ready for 3D printing (0.15 mm clearance and 3.2 mm bore already applied, with a bill of materials).



The save dialog: a real file explorer, with its shortcuts (home folder, Desktop, Documents, Pictures, Downloads), the folder tree, folder creation and the file type filter.

On Linux too, a real file explorer

On Linux, the "native" dialog often goes through a portal that reduces it to a plain list. The program therefore imposes Qt's full dialog there, shown above, with the usual places in the side bar and mounted media (USB sticks, disks). On Windows and macOS it uses the system's dialog, which is already complete.

4 What the dials show

4.1 The readings panel

The right-hand panel can be detached (it is a window in its own right) and shows:

Row	Meaning
Date	Gregorian date matching the crank's position
Egyptian calendar	month and day of the 365-day calendar on the front
Sun (zodiac)	position of the mean Sun — also the date pointer
Moon (zodiac)	position of the Moon, anomaly included
Anomaly correction	offset produced by the pin and slot, at most ±6.58°
Moon phase	name of the phase and illuminated percentage
Line of nodes	position of the nodes, which decides eclipses
Metonic dial	cell out of 235 and year out of 19 of the cycle
Callippic dial	quarter out of 4 of the 76-year cycle
Games dial	panhellenic games of the year
Saros spiral	cell out of 223
Exeligmos dial	time correction to add: 0, 8 or 16 hours
Moon and Sun	possible or near-certain eclipse warning
Planets (2021 model)	positions after the model of Freeth *et al.* (2021)
Mechanism vs real sky	**the instrument's own error**, in degrees

The last row is the most interesting one

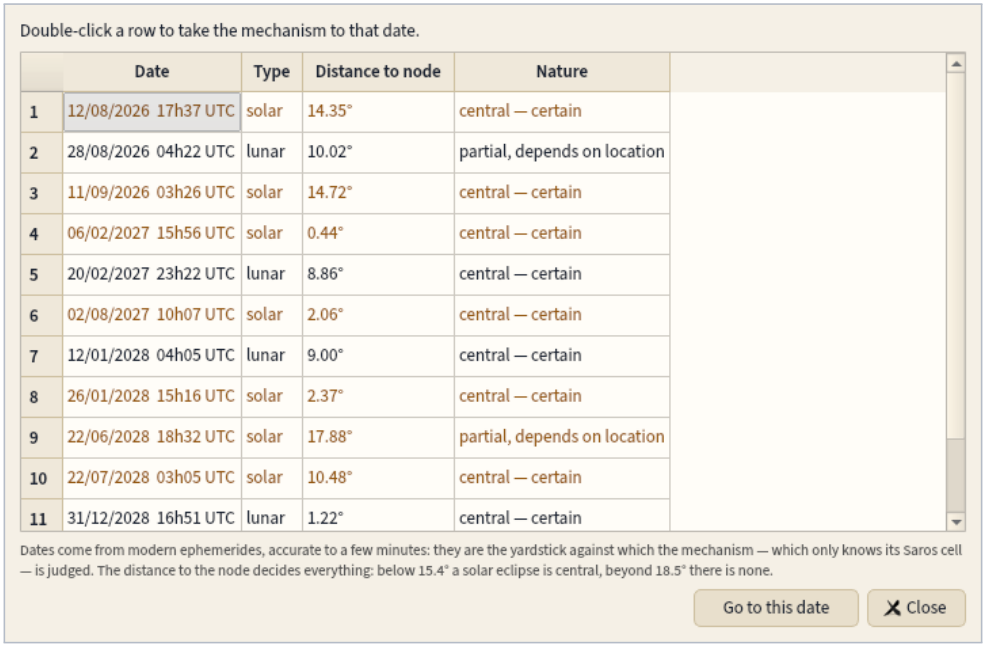
The program constantly compares what the machine shows with modern ephemerides. For the mean Sun the difference is **zero**: the ratio is exactly 1. For the Moon it reaches a few degrees per century, and that is not a software fault — it is the instrument's real error:

$$\underbrace{2.2^\circ}_{\substack{\text{evection and variation}\\ \text{not mechanised}}} + \underbrace{3.5^\circ \text{ per } 50 \text{ years}}_{\substack{\text{drift of}\\ 254/19}} + \underbrace{0.8^\circ}_{\substack{\text{precession of}\\ \text{the apogee}}}$$

The test suite checks that the measured error stays within this budget: the simulator is faithful *down to its flaws*.

4.2 Eclipses

Eclipses → **Upcoming eclipses…** (key E) lists the next twelve eclipses from the displayed date; a double-click, or **Go to this date**, takes the machine there.



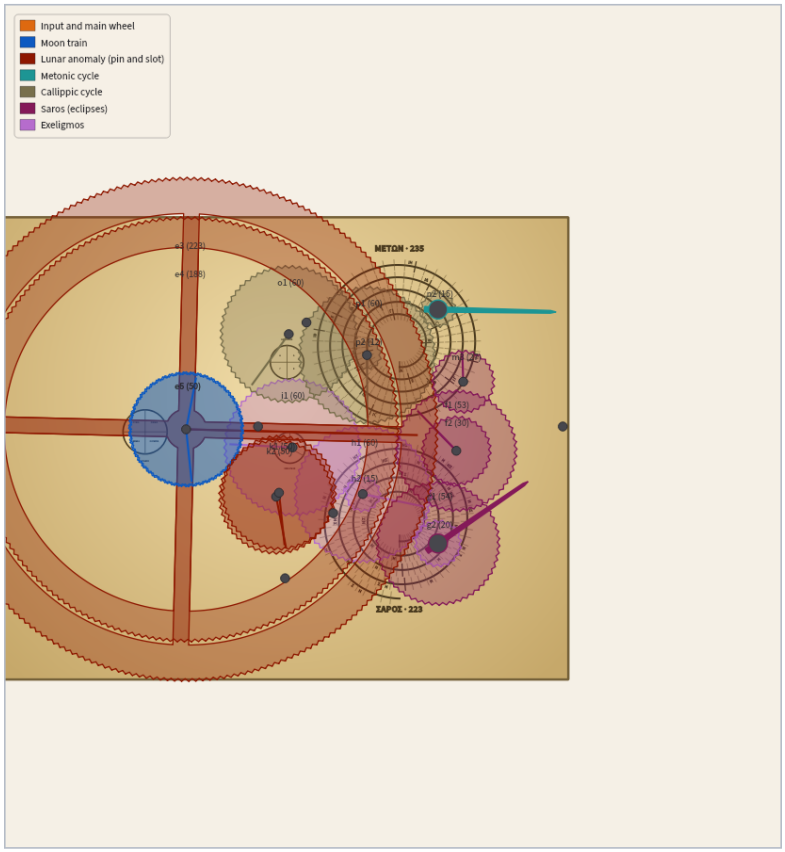
From 12 August 2026: the total solar eclipse of that very day (distance to node 14.35°), then the lunar eclipse of 28 August.

What the "Distance to node" column says

An eclipse is only possible at a new or full Moon close to a node of the lunar orbit. Below 15.4°, a solar eclipse is central somewhere on Earth; beyond 18.5° there is none. The dates in the list come from modern ephemerides, to within a few minutes: they are a reference for judging the machine, which itself only knows the Saros cell.

4.3 Reading the inscriptions

The dials carry inscriptions, reproduced from the fragments. Turn the machine over (key B) and zoom in: everything is engraved as vectors, sharp at any magnification.



The back face in vector rendering, mechanism showing through: Metonic spiral at the top, Saros spiral at the bottom, the great four-spoked wheel e_3 that carries the pin and slot.

Dial	Divisions	What is written
Front face		
Zodiac	$12 \times 30°$	the twelve signs in Greek from the vernal point; note *Chêlai*, "the claws", where we say Libra
Parapegma	letters	small index letters scattered in the zodiac, pointing to a star calendar engraved on the plate
Calendar	365 days	the Egyptian months in Greek letters; the ring is **movable**, shifted one notch every four years
Back face		
Metonic	spiral, 235 cells	one cell = one lunar month, with the name of its Corinthian month; the year number in Greek numerals marks each of the 19 years
Callippic	4 sectors	which of the four 76-year cycles is running
Games	4 sectors	*Olympia*, *Nemea*, *Isthmia*, *Pythia* — the only non-astronomical dial
Saros	spiral, 223 cells	the cells where an eclipse is possible carry H (eta, for *Hêlios*, the Sun) or Σ (sigma, for *Selênê*, the Moon)
Exeligmos	3 sectors	0, 8 or 16 hours to **add** to the time read on the Saros

The seven thirteen-month years

Nineteen solar years make 235 lunar months, but $19 \times 12 = 228$: seven years of the cycle have **thirteen** months. They are spread by the criterion $(12k) \bmod 19 < 7$, i.e. years $2, 5, 8, 10, 13, 16, 19$ — and the total comes back to exactly 235 cells. That is why the year-start marks are not evenly spaced on the spiral.

Why 223 cells on the Saros

Three lunar periods coincide there to within five hours over eighteen years: 223 synodic months (same phase), 242 draconic (same node), 239 anomalistic (same distance). The eclipse repeats. A remainder of 0.321 day shifts it by 7.7 hours — about 116° of longitude. After three Saros, that remainder is almost a whole day: this is what the Exeligmos dial counts.

Caveat: the eclipse glyphs are **computed**, not copied. The plate's pattern is known only from fragments; the program applies the criterion the machine mechanises — syzygy close to a node — over the 223 months of a Saros, and marks 48 solar cells and 32 lunar ones, 22 of which carry both.

5 The ratios, and why they are exact

Help → The mathematics of the mechanism gives the full table. Ratios are kept as **exact fractions**, never as decimal numbers:

Output	Ratio	Check
Sidereal Moon	254/19	sidereal month of 27.321 266 d
Carrier e_3	477/4237	8.8826 years (real apogee: 8.8504)
Anomaly	—	period of **27.5533 d** against 27.554 55: **1.8 minutes**
Metonic	5/19	5 turns in 19 years
Callippic	1/76	1 turn in 76 years
Saros	940/4237	4 turns for 223 lunations
Exeligmos	235/12711	1 turn in 54.09 years, i.e. 3 Saros

The anomalistic month is written nowhere

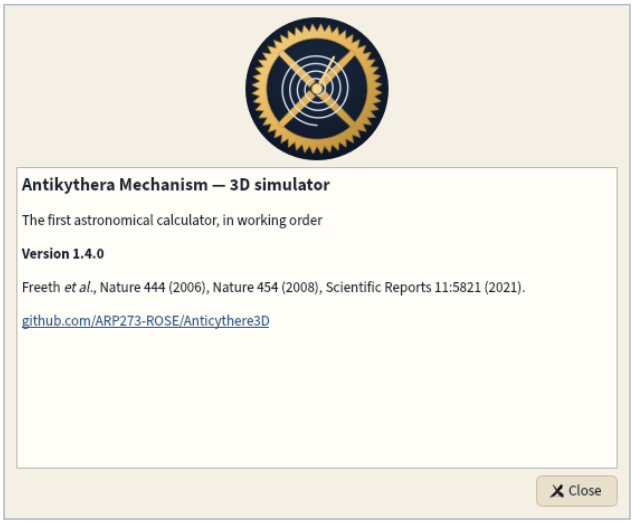
The pin and slot are mounted on wheel e_3, which turns in 8.88 years — the precession of the lunar apogee. The modulation's period is therefore the *difference* between the two rotations:

$$\frac{254}{19} - \frac{477}{4237} = \frac{56\,165}{4237}\ \text{turn/year} \quad \Longrightarrow \quad T = 27.553\ \text{days},$$

to be compared with the real anomalistic month of 27.554 55 days: **1.8 minutes apart**. This number appears on no wheel: it emerges from a subtraction of frequencies made in bronze.

6 The Help menu

Entry		Effect
Manual (PDF)	F1	opens this manual, in the interface language
Check for updates…		queries the published releases
Report a problem…		sends a written report
Send crash reports automatically		consent given or withdrawn
Quick guide		built-in summary, with the colour legend
The mathematics of the mechanism		table of the exact ratios
Keyboard shortcuts		every shortcut and gesture
About		version, sources, and the program's logo



The "About" window: the logo is the same as the icon's, the installer's and this manual's — one single drawing, the four-spoked great wheel b_1 and the Saros spiral.

7 Troubleshooting

Symptom	What to do
"Missing dependencies" (source)	run `python run.py` again, or `pip install -r requirements.txt`
Black window, no 3D	OpenGL unavailable: switch to **vector rendering** (the program does it by itself)
Jagged teeth	raise the antialiasing: `--samples 16`, or use vector rendering
Animation too slow	lower the speed, untick **Gear labels**
Only the large wheels are visible	expected: b_1 and e_3 cover the small ones. Use **Level explosion** or **Highlight**
The crank cannot be grabbed	aim at the handle itself: the cursor must become a hand
The view is tilted or upside down	**Reset** or R
"Windows protected your PC"	unsigned installer: **More info** → **Run anyway**
macOS refuses to open the application	right-click → **Open**, or System Settings → Privacy & Security → Open Anyway
The update fails	close the other copy of the program, then ask again

8 What is faithful, and what is not

Faithful and checked

The 30 attested gears and their tooth counts, the ratios of every train, the pin-and-slot device and its carrier, the back dials (Metonic spiral of 5 turns and 235 cells, Saros spiral of 4 turns and 223 cells), the principle of the single crank, the four spokes of b_1.

Reconstructed or simplified

- The **planets** are computed after the 2021 model, but not modelled in 3D: they are reconstructions, not surveys.
- The **layout of the arbors** is computed by constrained optimisation; it keeps the 17 exact centre distances, but it is not the original's arrangement, which remains partly unknown.
- The Saros **eclipse glyphs** are computed, not copied from the plate, known only from fragments.
- The **order of the levels** is the project's, not a survey: k_1, k_2 are carried by e_3 as in the reconstructions by Freeth *et al.*, but the stacking of the other trains is chosen so that no arbor goes through a gear.
- The **case** follows the proportions of the reconstructed mechanism, not the original's $34 \times 18 \times 9$ cm.

Sources. Freeth T. *et al.*, *Nature* **444** (2006) 587–591; Freeth T., Jones A., Steele J., Bitsakis Y., *Nature* **454** (2008) 614–617; Freeth T., Higgon D., Dacanalis A., MacDonald L., Georgakopoulou M., Wojcik A., *Scientific Reports* **11**:5821 (2021); Meeus J., *Astronomical Algorithms*, 2nd ed.